

PARKTOWN BOYS' HIGH SCHOOL



Department of Mathematics

June Examination Paper 2

5 June 2023

Subject	Mathematics
Grade	12
Marks	150
Time	3 hours

Examiner

Mrs H Gouws

Moderator

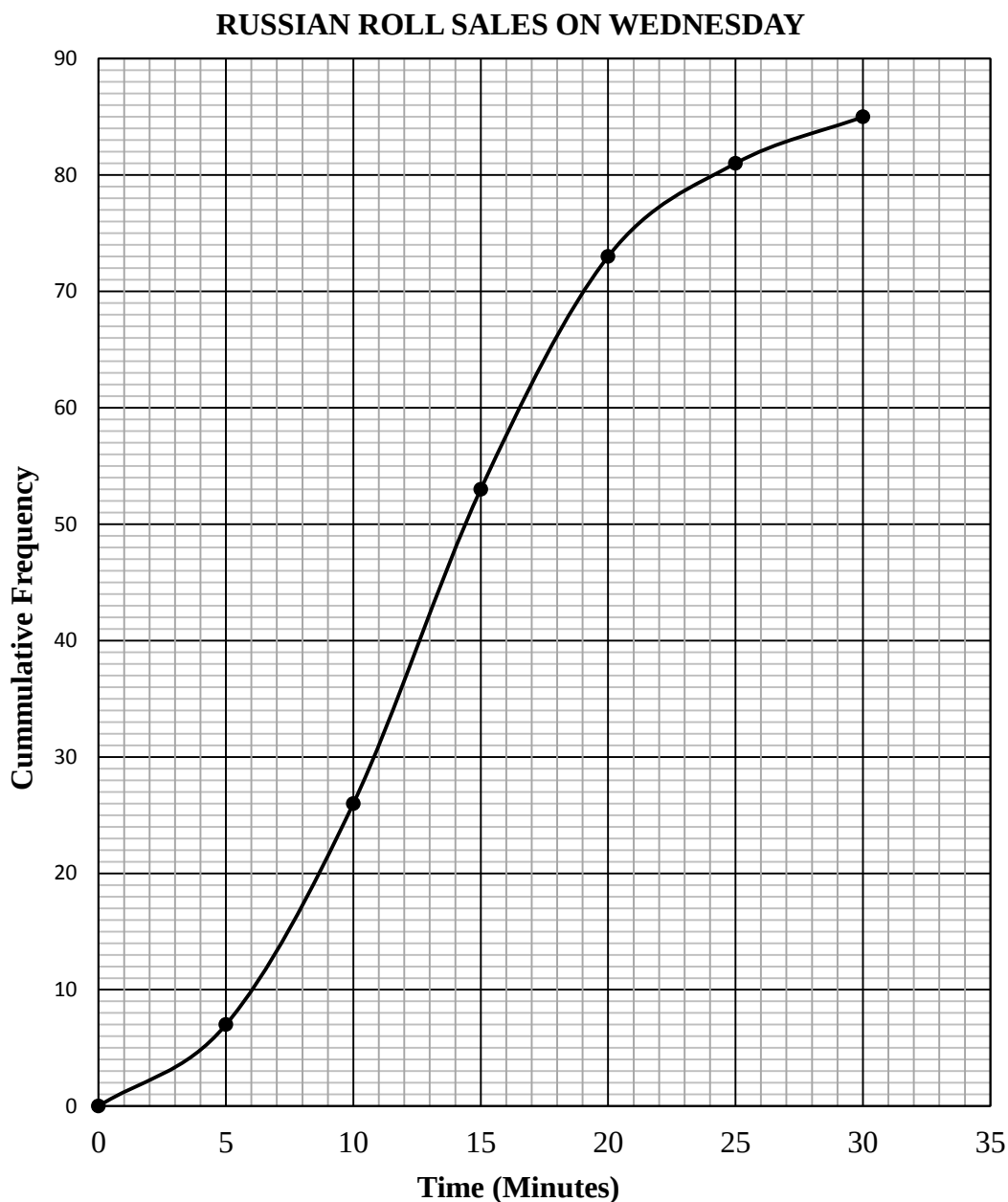
Mrs S Rowell

INSTRUCTIONS AND INFORMATION

1. This paper consists of 12 pages and a formula sheet.
 - You may detach the formula sheet for your own use.
2. There are 11 questions.
3. Answer all question in the answer book provided. Additional space is provided at the end of the answer book.
4. Round all answers off to TWO decimals unless indicated otherwise in a particular question.
5. Approved non-graphical and non-programmable calculators may be used.
6. An answer alone will not necessarily be awarded full marks.
7. Look at the mark allocation as an indication of the length of the solution.
8. Write neatly and legibly.

QUESTION 1

The cumulative frequency graph below shows the total number of Russian Rolls sold during the 30-minute break on a Wednesday.



- 1.1 How many Russian Rolls did Uncle Joe sell in total during this break? (1)
- 1.2 What is the modal class? (1)
- 1.3 How many Russian Rolls were bought during the last 12 minutes of break? (2)
- 1.4 Use the graph to determine the median of this data set. Clearly indicate how you found this value on the graph in the answer book. (2)

- 1.5 At the end of the day Uncle Joe realises that four Russian Rolls were sold in the last five minutes of break and not counted.

- 1.5.1 Complete the table given below in your answer book with these four Russian Rolls included.

Class	Frequency
$0 \leq x < 5$	
$5 \leq x < 10$	
$10 \leq x < 15$	
$15 \leq x < 20$	
$20 \leq x < 25$	
$25 \leq x < 30$	

(2)

- 1.5.2 Calculate the estimated mean of this data.

(3)

- 1.5.3 Comment on the distribution of the data.

(1)

[12]

QUESTION 2

Modern Lawn Services offers lawn care services, including mowing of lawns to customers. To calculate the charges for their lawn mowing services, they measure the approximate area, x (in m^2) of a random sample of 12 customers' lawns and the time, y (in minutes) that it takes them to mow the lawns.

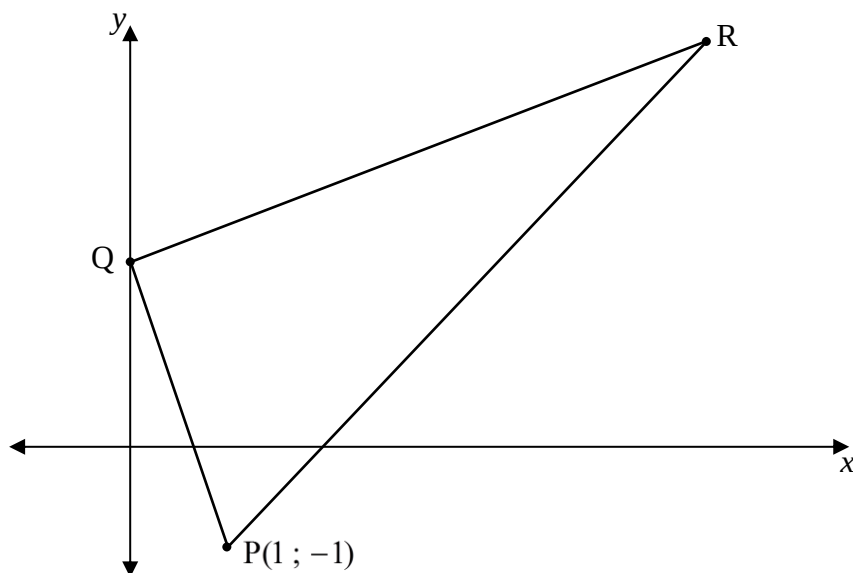
Area in m^2 (x)	360	120	845	602	1190	530	245	486	350	1005	320	250
Time in minutes (y)	50	28	130	75	120	95	55	70	48	110	55	60

- 2.1 Determine the equation of the least squares regression line. (2)
- 2.2 Sketch the least squares regression line on the scatterplot provided in the answer book. (2)
- 2.3 Calculate the correlation coefficient. (1)
- 2.4 Comment on the relationship between the area to be mowed and the time taken. (3)
- 2.5 Modern Lawn Services charge a flat call-out fee of R200, as well as R100 per half hour (or part thereof), estimate the charge for mowing a customer's lawn that has an area of 810 m^2 . (3)
(For example: 130 minutes would be taken as $2\frac{1}{2}$ hours)
- 2.6 Parktown Boys' High School approaches Modern Lawn Services to mow the top rugby field which is 100 m long and 70 m wide. Should the equation of the regression line found in QUESTION 2.1 be used to estimate the time it would take to mow the area? Give a reason for your answer. (2)

[13]

QUESTION 3

In the diagram below, PQR is a triangle. The coordinates of P are $(1; -1)$. The equations of QR and PR are $y = \frac{1}{3}x + 2$ and $x - y - 2 = 0$ respectively.

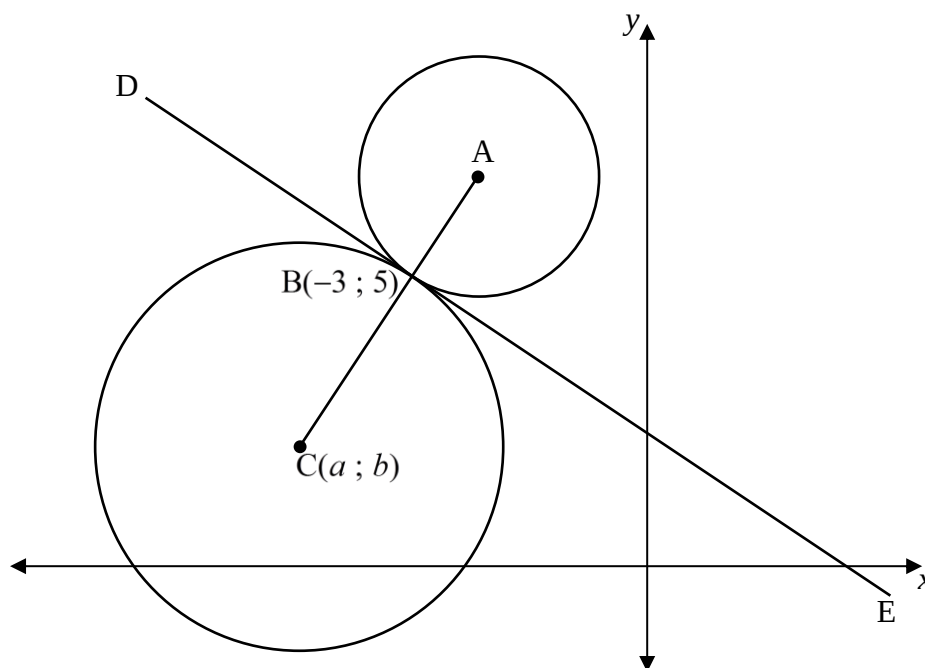


- 3.1 Determine the coordinates of Q. (2)
- 3.2 Prove that $\hat{PQR} = 90^\circ$. (3)
- 3.3 Determine the coordinates of R. (4)
- 3.4 Determine the coordinates of T if PQRT is a parallelogram. (2)
- 3.5 Determine the size of $\hat{QRP}$. (4)
- 3.6 Determine the equation of the circle through the points P, Q and R. (5)

[20]

QUESTION 4

In the diagram below, the circles with centres A and C are drawn. The equation of the circle with centre A is $(x+2)^2 + (y-6)^2 = r^2$. DE is a common tangent to circles A and C at B(-3 ; 5).



- 4.1 Write down the coordinates of A. (2)
- 4.2 Determine the radius of circle A. (2)
- 4.3 Determine the equation of the tangent DE. (4)
- 4.4 If $\frac{AB}{BC} = \frac{1}{3}$, determine the coordinates of C. (8)

[16]

QUESTION 5

- 5.1 If $\cos 2A = -\frac{3}{5}$ where $0^\circ \leq 2A \leq 180^\circ$, determine the value of each of the following trigonometric ratios without the use of a calculator and with the aid of a diagram:
- 5.1.1 $\sin 2A$ (4)
- 5.1.2 $\cos A$ (3)

5.2 Simplify: $\frac{\sin(\theta - 180^\circ) \cdot \tan(-\theta) \cdot \cos(360^\circ - \theta)}{\cos^2(90^\circ + \theta) + \sin^2(180^\circ + \theta)}$ (6)

5.3 Calculate $\frac{(\cos 15^\circ + \sin 15^\circ)^2}{\cos^2 15^\circ - \sin^2 15^\circ}$ without the use of a calculator. (5)

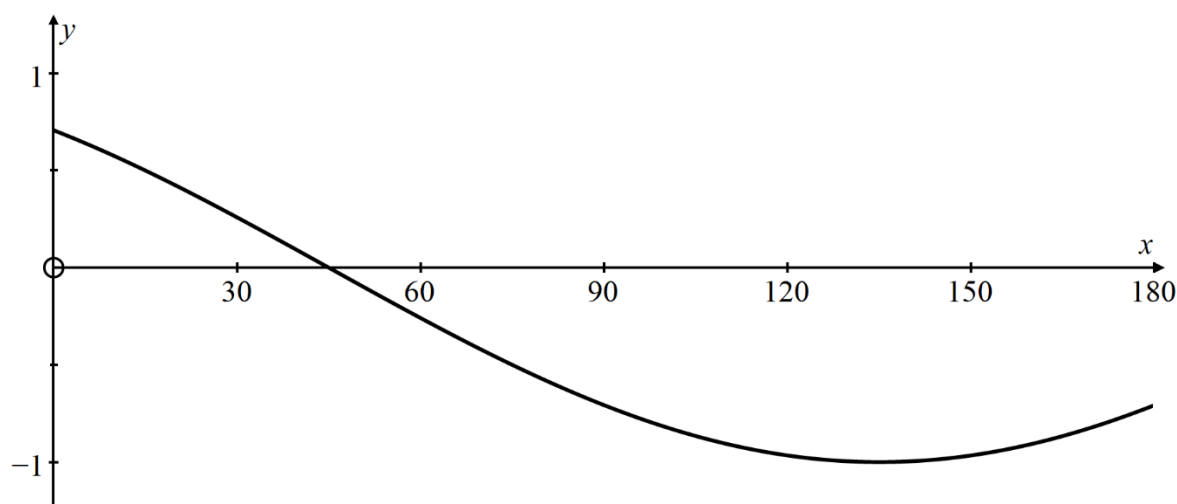
5.4 Prove the identity $\frac{\cos 2x - \cos x - 2}{\sin 2x - 3\sin x} = \frac{\cos x + 1}{\sin x}$. (5)

5.5 If $\sin 28^\circ = p$ and $\cos 32^\circ = q$, prove without the use of a calculator that $q\sqrt{1-p^2} - p\sqrt{1-q^2} = \frac{1}{2}$. (5)

[28]

QUESTION 6

In the diagram below, the graph of $f(x) = \cos(x + 45^\circ)$ for $x \in [0^\circ; 180^\circ]$ has been sketched:



6.1 On the set of axes provided in the answer book sketch the graph of $g(x) = \sin 3x$ for $x \in [0^\circ; 180^\circ]$. Show all relevant details such as turning points and intercepts with the axis on your sketch. (3)

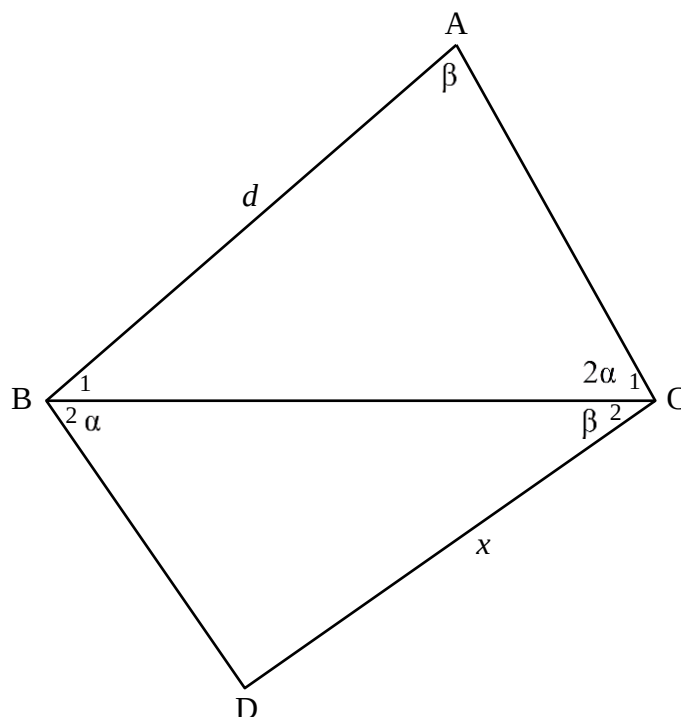
6.2 What is the period of g ? (1)

6.3 Determine the value(s) of $x \in [0^\circ; 180^\circ]$ for which $f(x) = g(x)$. (6)

[10]

QUESTION 7

In the diagram below, $\hat{A} = \hat{C}_2 = \beta$, $\hat{B}_2 = \alpha$, $\hat{C}_1 = 2\alpha$, $CD = x$ and $AB = d$:



7.1 Determine the size of $\hat{D}$ in terms of α and β . (2)

7.2 Determine the length of BC in terms of α , β and x . (4)

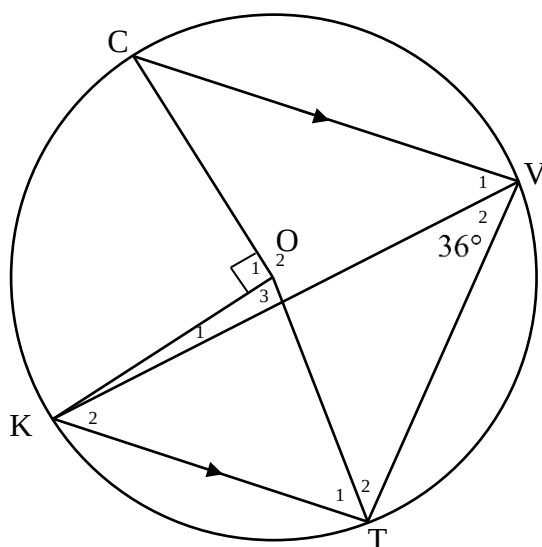
7.3 Prove that $d = \frac{2x \sin(\alpha + \beta) \cos \alpha}{\sin \beta}$. (4)

7.4 If $\alpha = 30^\circ$, $\beta = 45^\circ$ and $x = 100$ m determine d . (2)

[12]

QUESTION 8

In the diagram below, O is the centre of the circle. $CV \parallel KT$, $CO \perp OK$ and $\hat{V}_2 = 36^\circ$.



Determine the size of

8.1 $\hat{V}_1$ (2)

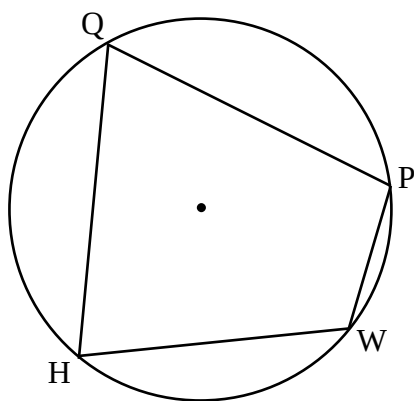
8.2 $\hat{O}_2$ (that is reflex $\hat{COT}$) (3)

8.3 $\hat{K}_1$ (4)

[9]

QUESTION 9

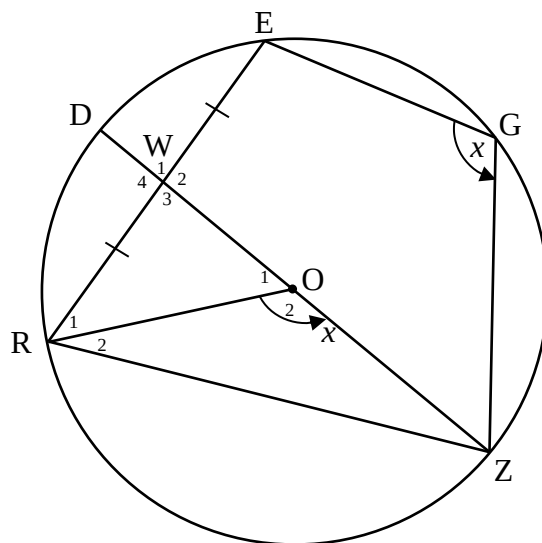
9.1 In the diagram below, O is the centre of the circle and QPWH is a cyclic quadrilateral.



Use the diagram in the answer book to prove the theorem which states that $\hat{Q} + \hat{W} = 180^\circ$.

(5)

- 9.2 In the diagram below, O is the centre of the circle. DZ bisects RE at W.
 $\hat{G} = \hat{O}_2 = x$.

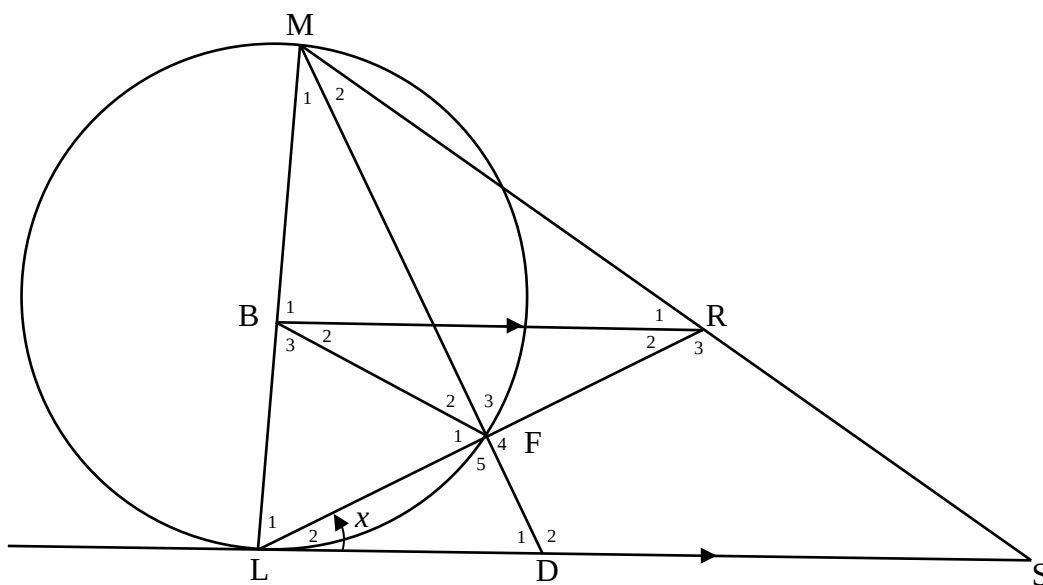


Determine

- 9.2.1 the size of $\hat{W}_3$. (1)
- 9.2.2 the size of $\hat{ERZ}$ in terms of x . (2)
- 9.2.3 the value of x . (5)

[13]

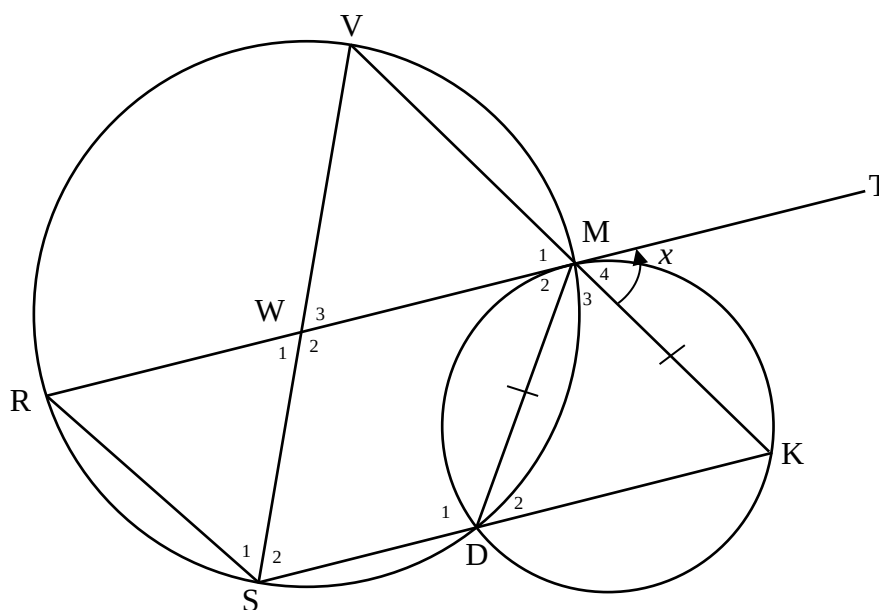
In the diagram below, LS is a tangent to the circle at L. $BR \parallel LS$ and MD bisects $\angle LMR$. $\angle SLR = x$.



- [10]**

QUESTION 11

In the diagram below, RT is a tangent to the smaller circle at M . $\hat{TMK} = x$ and $MD = MK$.



Prove that

11.1 $RT \parallel SK$. (3)

11.2 MD is a tangent to the circle through VMW at M . (4)

[7]